Multiple Linear Regression Using R
Lab Session-11
Course Code : MAT445

Date:15.10.2026 **Time : 3:15PM - 4:15 PM**

1 Aim

To construct and evaluate a multiple linear regression model using a CSV dataset in R. The experiment includes data exploration, coefficient estimation, prediction, model evaluation, regression diagnostics, and multicollinearity analysis.

2 Problem Statement

Read the file `data.csv` and construct a multiple linear regression model to predict final examination marks using study hours, attendance, and previous examination marks.

Perform descriptive analysis, fit the regression model, interpret the results, predict the marks of a new student, evaluate predictive performance using error metrics, and calculate the Variance Inflation Factor (VIF).

3 Mathematical Model

The multiple linear regression model is

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \varepsilon_i$$

where

- Y_i represents final examination marks.
- X_{i1} represents study hours.
- X_{i2} represents attendance.
- X_{i3} represents previous examination marks.
- $\beta_0, \beta_1, \beta_2, \beta_3$ are regression coefficients.
- ε_i represents the random error.

4 Complete R Program

4.1 Step 1: Import and Explore the Dataset

```r
# Read the CSV file
data <- read.csv(
  "C:/Users/senth/OneDrive/Desktop/data.csv",
  header = TRUE
)

# Display the dataset
print(data)

# Display the first six rows
head(data)

# Examine data structure
str(data)

# Summary statistics
summary(data)

# Dimensions and column names
dim(data)
names(data)

# Count missing values in each column
colSums(is.na(data))
```

4.2 Step 2: Correlation and Scatterplot

```r
# Select the variables used in regression
variables <- c(
  "StudyHours",
  "Attendance",
  "PreviousMarks",
  "FinalMarks"
)

# Correlation matrix
cor(
  data[, variables],
  use = "pairwise.complete.obs"
)

# Scatterplot
plot(
  data$StudyHours,
  data$FinalMarks,
  main = "Study Hours vs Final Marks",
```

```
  xlab = "Study Hours",
  ylab = "Final Marks",
  pch = 19
)
```

4.3 Step 3: Prepare the Modeling Data

```
# Select required columns and remove incomplete rows
model_data <- na.omit(data[, variables])

# Display the number of usable observations
cat("Usable observations:",
    nrow(model_data), "\n")
```

4.4 Step 4: Fit Multiple Linear Regression

```
# Fit the regression model
model <- lm(
  FinalMarks ~ StudyHours +
    Attendance + PreviousMarks,
  data = model_data
)

# Complete regression summary
summary(model)

# Estimated regression coefficients
coef(model)

# R-squared
summary(model)$r.squared

# Adjusted R-squared
summary(model)$adj.r.squared

# Overall F-statistic
summary(model)$fstatistic

# Confidence intervals
confint(model)

# Analysis of variance table
anova(model)
```

4.5 Step 5: Predict and Compare Observations

```
# Predictions on the same data used for fitting
```

```r
predicted <- predict(model)

# Actual values, predictions, and residuals
results <- data.frame(
  Actual = model_data$FinalMarks,
  Predicted = predicted,
  Residual = residuals(model)
)

print(results)

# Predict marks for a new student
new_student <- data.frame(
  StudyHours = 6,
  Attendance = 85,
  PreviousMarks = 75
)

prediction <- predict(
  model,
  newdata = new_student
)

cat("Predicted final marks:",
    prediction, "\n")
```

4.6 Step 6: Calculate Error Metrics

```r
actual <- model_data$FinalMarks
predicted <- predict(model)

# Mean Absolute Error
MAE <- mean(abs(actual - predicted))

# Root Mean Squared Error
RMSE <- sqrt(mean((actual - predicted)^2))

# Mean Absolute Percentage Error
# MAPE requires nonzero actual values
if (all(actual != 0)) {
  MAPE <- mean(
    abs((actual - predicted) / actual)
  ) * 100

  cat("MAPE =", MAPE, "%\n")
} else {
  cat("MAPE cannot be calculated when actual",
      "values contain zero.\n")
}

```

```r
cat("Training MAE =", MAE, "\n")
cat("Training RMSE =", RMSE, "\n")
```

4.7 Step 7: Split into Training and Testing Sets

```r
set.seed(123)

# Select 70 percent for training
n <- nrow(model_data)

train_index <- sample(
  seq_len(n),
  size = floor(0.7 * n)
)

train_data <- model_data[train_index, ]
test_data <- model_data[-train_index, ]

# Fit the model using training observations
model_train <- lm(
  FinalMarks ~ StudyHours +
    Attendance + PreviousMarks,
  data = train_data
)

# Predict on test observations
test_predictions <- predict(
  model_train,
  newdata = test_data
)

# Test-set evaluation
test_MAE <- mean(
  abs(test_data$FinalMarks - test_predictions)
)

test_RMSE <- sqrt(
  mean(
    (test_data$FinalMarks - test_predictions)^2
  )
)

cat("Test MAE =", test_MAE, "\n")
cat("Test RMSE =", test_RMSE, "\n")
```

4.8 Step 8: Regression Diagnostic Plots

```r
# Four standard diagnostic plots
par(mfrow = c(2, 2))
```

```r

plot(model_train)

# Restore the default plotting layout
par(mfrow = c(1, 1))
```

4.9 Step 9: Calculate VIF Without Additional Packages

The Variance Inflation Factor measures how strongly each predictor can be explained by the remaining predictors.

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

Here, R_j^2 is obtained by regressing predictor j on all the other predictors.

```r
# Select independent variables from training data
X <- train_data[, c(
  "StudyHours",
  "Attendance",
  "PreviousMarks"
)]

# Calculate VIF for every predictor
for (variable in names(X)) {

  # Select all other predictors
  other_variables <- X[
    ,
    names(X) != variable,
    drop = FALSE
  ]

  # Auxiliary regression
  auxiliary_model <- lm(
    X[[variable]] ~ .,
    data = other_variables
  )

  # R-squared of the auxiliary regression
  R2 <- summary(auxiliary_model)$r.squared

  # Guard against division by zero
  if (1 - R2 < 1e-10) {
    VIF <- Inf
  } else {
    VIF <- 1 / (1 - R2)
  }

  cat(
    variable,
```

```
    "VIF =",
    round(VIF, 3),
    "\n"
  )
}

# Correlation among the independent variables
cor(X)
```

5 Interpretation of Results

1. **Regression coefficients:** Describe the estimated change in final marks for a one-unit increase in a predictor, holding the other predictors constant.
2. R^2**:** Indicates the proportion of variation in final marks explained by the fitted model on the observations used for fitting.
3. **Adjusted** R^2**:** Adjusts the coefficient of determination for the number of predictors and sample size.
4. **MAE and RMSE:** Measure prediction errors. Lower values indicate smaller errors on the evaluated observations.
5. **Diagnostic plots:** Help investigate linearity, residual variance, residual normality, and influential observations.
6. **VIF:** A large value indicates substantial linear dependence between a predictor and the remaining predictors. Values above 5 or 10 are common warning thresholds, not universal decision rules.

6 Important Notes

- Training-set error is not a reliable estimate of performance on unseen data.
- A small test dataset can produce unstable evaluation metrics. Use more observations or cross-validation when appropriate.
- High VIF does not automatically imply poor predictive accuracy. It primarily complicates the interpretation and estimation of individual regression coefficients.
- If an auxiliary regression has $R_j^2 = 1$, the corresponding VIF is infinite. This indicates perfect linear dependence in that sample.
- The CSV file must contain the four named columns used in this experiment.

7 Result

A multiple linear regression model is constructed using R. Its coefficients, explanatory power, predictions, prediction errors, diagnostic plots, and multicollinearity are examined.